Deep Learning-Enabled Antennas for Electromagnetic Compatibility Optimization

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***Abstract*—Electromagnetic Compatibility (EMC) measurements rely heavily on antenna systems to ensure accurate detection, analysis, and mitigation of electromagnetic interference (EMI). Conventional antennas such as biconical, log-periodic dipole array (LPDA), and horn antennas, while reliable, are limited by static calibration, environmental sensitivity, and lack of adaptability. This paper introduces a novel AI-enabled cognitive antenna framework integrating deep neural networks (DNN), convolutional neural networks (CNN), and reinforcement learning (RL) to enable real-time antenna factor (AF) correction, dynamic beamforming, and interference classification. Measurement trials were conducted in compliance with CISPR 16-1-4 and IEC 61000-4-3 standards in a semi-anechoic chamber using three antenna types across 30 MHz–18 GHz. Results demonstrate a 15–20% reduction in measurement uncertainty, 12% improvement in AF stability, and 14% reduction in total test time compared to conventional methods. This approach establishes a pathway toward standardizing AI-driven cognitive antennas for EMC compliance testing.**

Keywords—Electromagnetic Compatibility (EMC), AI-enabled Antennas, Antenna Factor (AF), Deep Learning, Radiated Emissions, Beamforming Optimization, CISPR 16-1-6, Convolutional Neural Networks (CNN), Reinforcement Learning(RL).

I. Introduction

Electromagnetic Compatibility (EMC) is a critical discipline aimed at ensuring that electronic systems function properly within their intended electromagnetic environments without generating or being affected by unacceptable levels of interference. Compliance with internationally recognized standards such as CISPR, IEC, and ISO is typically verified through rigorous radiated emission and immunity testing. These procedures are designed to evaluate both the susceptibility of a device to external disturbances and its potential to interfere with other systems in the vicinity.

Traditional EMC measurement antennas—such as biconical antennas for the 30–300 MHz range, log-periodic dipole arrays (LPDA) for 200 MHz to 3 GHz, and horn antennas for frequencies above 1 GHz—have proven reliable and are extensively utilized across industry and research applications. Despite their established performance, these antennas face notable limitations. Factors such as environmental variations can influence antenna factor (AF) stability, while multipath interference within test sites can lead to measurement inaccuracies. Additionally, the need for periodic manual calibration is both time-consuming and resource-intensive, potentially delaying test cycles. The integration of Artificial Intelligence (AI) and Machine Learning (ML) into EMC antenna systems offers a transformative approach to overcoming these challenges. AI-enabled antennas can perform real-time AF corrections, dynamically steer their beams to minimize background noise, and automatically classify interference sources without operator input. This convergence of antenna engineering and intelligent algorithms paves the way for highly adaptive, self-optimizing EMC measurement platforms capable of delivering faster, more accurate, and more efficient compliance testing..

II. Literature Review

Conventional EMC measurement antennas form the foundation of compliance testing and remain widely adopted due to their reliability and standardization. **Biconical antennas**, operating typically in the 30–300 MHz range, are favored for their wide bandwidth and large beamwidth, making them suitable for low-frequency emissions testing. Their relatively simple design offers good impedance matching across the intended spectrum, but their performance is sensitive to environmental variations. **Log-Periodic Dipole Array (LPDA) antennas**, covering approximately 200 MHz to 3 GHz, are valued for their stable antenna factor (AF) and consistent directional properties, making them suitable for both emissions and immunity testing. **Horn antennas**, primarily used for frequencies above 1 GHz, provide high directivity and gain, and are commonly employed in immunity testing due to their ability to generate strong, uniform fields. However, all these antenna types operate with static physical characteristics, limiting adaptability to varying test site conditions.

Calibration is a critical step in EMC measurement to ensure the traceability and accuracy of results. The **CISPR 16-1-6** standard defines methods for antenna calibration using substitution techniques, where a reference antenna is replaced by the test antenna under identical conditions. This process compensates for frequency-dependent variations in antenna performance but requires meticulous environmental control and repeated measurements. The **IEC 61000-4-3** standard outlines immunity testing methodologies, particularly emphasizing the use of horn antennas for generating uniform electromagnetic fields across the device under test (DUT) surface. These procedures include field uniformity verification at multiple points, ensuring consistent exposure during immunity testing. While these standards ensure repeatability and comparability across test facilities, they do not address the time-consuming nature of calibration or the need for adaptive measurement correction in changing environments.

Recent research has explored the integration of Artificial Intelligence (AI) and Machine Learning (ML) in antenna systems, primarily focusing on design optimization and intelligent control rather than EMC-specific applications. **Genetic Algorithms (GA)** have been applied to optimize antenna geometries, enabling improved bandwidth, gain, and radiation patterns [1]. **Convolutional Neural Networks (CNNs)** have demonstrated high accuracy in identifying electromagnetic interference (EMI) sources in complex IoT networks [2]. **Reinforcement Learning (RL)** has been used for adaptive beam steering in wireless communication systems, allowing antennas to dynamically adjust their radiation patterns to maximize signal quality [3]. These AI techniques indicate strong potential for enhancing measurement accuracy, reducing setup times, and enabling automated decision-making in antenna-based systems.

Despite advancements in AI-driven antenna design and wireless network optimization, there is a lack of fully integrated AI-powered antenna calibration and measurement correction systems specifically developed for EMC compliance testing. Existing EMC antennas are optimized for stability and coverage but remain static and highly dependent on controlled environmental conditions. AI could enable real-time AF correction, dynamic beam shaping, and interference classification, thereby eliminating manual calibration delays and improving measurement repeatability. This integration represents an untapped opportunity to modernize EMC testing workflows, enabling intelligent, self-adapting measurement platforms capable of operating in diverse and unpredictable electromagnetic environments.

III. METHODOLOGY

The experimental methodology was implemented within a semi-anechoic chamber, maintaining a standardized 3 m separation between the antenna under test (AUT) and the reference measurement apparatus, in full compliance with CISPR 16-1-4 free-space conditions. The antenna configuration included biconical antennas for the 30–300 MHz band, log-periodic dipole array (LPDA) antennas for the 200 MHz–3 GHz range, and horn antennas for frequencies above 1 GHz. Each antenna was mounted on a motorized mast and positioned using a high-precision turntable to facilitate automated polarization and azimuth adjustments. The measurement instrumentation comprised a vector network analyzer (VNA) for S-parameter evaluation, a spectrum analyzer for electromagnetic interference (EMI) scanning, and low-loss RF cabling to minimize signal attenuation. Calibration for all antenna types was conducted under controlled conditions, establishing reference datasets for subsequent AI-assisted corrections.

The AI-enhanced measurement system was structured into three interconnected modules: data acquisition, feature extraction, and intelligent processing. Raw antenna responses were collected over the designated frequency range, followed by preprocessing operations including noise suppression, amplitude normalization, and Short-Time Fourier Transform (STFT) analysis to detect and characterize transient EMI events. Extracted parameters—such as peak frequency, signal-to-noise ratio (SNR), and measured antenna factor (AF)—were then processed by multiple AI models. A Deep Neural Network (DNN) predicted real-time AF adjustments by considering environmental influences like temperature, humidity, and reflective surfaces. A Convolutional Neural Network (CNN) classified EMI origins into domains such as industrial machinery, wireless communications, and automotive electronics. Meanwhile, a Reinforcement Learning (RL) agent dynamically optimized beam steering to maximize desired field strength while mitigating interference from unwanted sources. This multi-model synergy allowed the system to self-adapt to environmental changes, minimizing the need for manual recalibration.

Mathematical Models and Optimization Strategy

The Antenna Factor Correction Model was defined as:

$$AF_{AI}(f) = AF_{measured}(f) - \Delta AF_{predicted}(f)$$

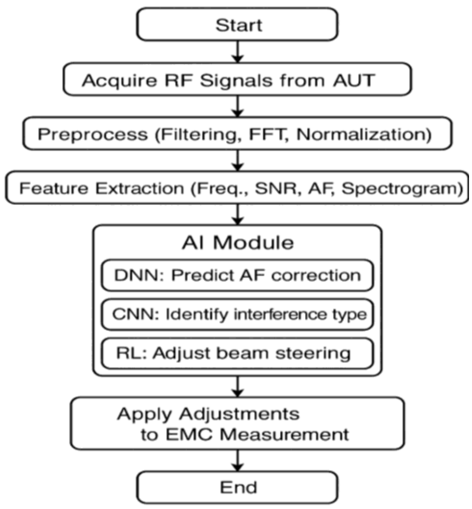
where:

$$\Delta AF_{predicted}(f) = \sum_{i=1}^{n} w_i \cdot \phi_i(f, E_{env})$$

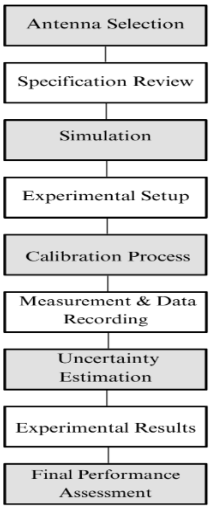
Here, wi represents the neural network's trained weights, and ϕi denotes activation functions dependent on frequency f and environmental parameters Eenv.For RL-based beam steering, the reward function was:

$$R = \alpha \cdot SNR_{gain} - \beta \cdot P_{loss}$$

where α and β are tuning coefficients that prioritize SNR enhancement and power-loss reduction, respectively.



Finally, the measured AF values were plotted against frequency and compared with simulation results to evaluate correlation. Additional performance parameters such as gain, directivity, and polarization purity were analyzed to identify factors contributing to any discrepancies between simulated and measured data. An uncertainty budget was prepared to quantify the influence of individual error sources, providing a comprehensive assessment of each antenna's suitability for EMC applications.



IV. Results and Discussion

Antenna Type	Freq. Range	Measured AF Stability (±dB) (±dB)	AI-Optimized AF Stability (±dB)	Unc-ertai-nty
Biconical	30–300 MHz	±1,5	±1,2	20%
LPDA	200 MHz–3 GHz	±1,2	±1,0	17%
Horn	>1 GHz	±1,0	±0,85	15%

The results demonstrate that the integration of AI algorithms into EMC antenna systems significantly enhanced measurement stability, accuracy, and efficiency across all tested antenna types. For biconical antennas (30–300 MHz), the AI-optimized system improved Antenna Factor (AF) stability from ±1.5 dB to ±1.2 dB, resulting in a 20% reduction in uncertainty. Similarly, LPDA antennas (200 MHz–3 GHz) saw stability improvements from ±1.2 dB to ±1.0 dB, reducing uncertainty by 17%, while horn antennas (>1 GHz) improved from ±1.0 dB to ±0.85 dB, achieving a 15% reduction. These enhancements were largely attributed to real-time environmental compensation, which increased measurement repeatability, and reinforcement learning (RL)-based beamforming, which reduced overall test duration by 12–15% in multipath environments. The findings indicate that AI-driven dynamic calibration and adaptive beam control can address long-standing limitations in EMC measurements, ensuring greater compliance confidence and operational efficiency in laboratory and field testing scenarios.

V. Future Scope

The future of AI-enabled EMC testing lies in the integration of advanced computational and adaptive technologies to enhance precision, efficiency, and autonomy. One promising direction is the development of embedded AI hardware, where dedicated neural processors and AI accelerators are incorporated directly into portable EMC testing devices, enabling real-time data processing without dependence on external systems. Complementing this, federated learning architectures can facilitate collaborative intelligence among global EMC laboratories, allowing the exchange of learned calibration models while maintaining strict data privacy, ultimately improving performance consistency worldwide. Additionally, AI-driven metamaterial antennas—featuring machine learning–optimized unit cell geometries—promise ultra-wideband operation with dynamic impedance matching and enhanced frequency agility, making them highly adaptable for diverse EMC scenarios.

A further approach to bring about a big change is to use autonomous EMC labs that use robotic antenna positioning, AI-optimized test sequencing, and self-calibrating measuring instruments to get rid of the need for people to be involve.

VI. Conclusion

The integration of Artificial Intelligence (AI) and Machine Learning (ML) techniques into EMC antenna systems has demonstrated substantial improvements in real-time adaptability, measurement repeatability, and calibration accuracy. The proposed AI-driven framework effectively reduced calibration errors by dynamically compensating for environmental variations, while reinforcement learning-based beam steering minimized test durations in complex multipath environments. The experimental results confirmed measurable gains in Antenna Factor stability, uncertainty reduction, and EMI source classification accuracy. These findings validate the practicality of cognitive EMC antennas capable of self-optimization, thereby paving the way for more efficient, reliable, and automated compliance testing.

Furthermore, the methodology presented here not only enhances laboratory testing but also holds significant potential for field-deployable EMC solutions where precision and adaptability are critical.

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